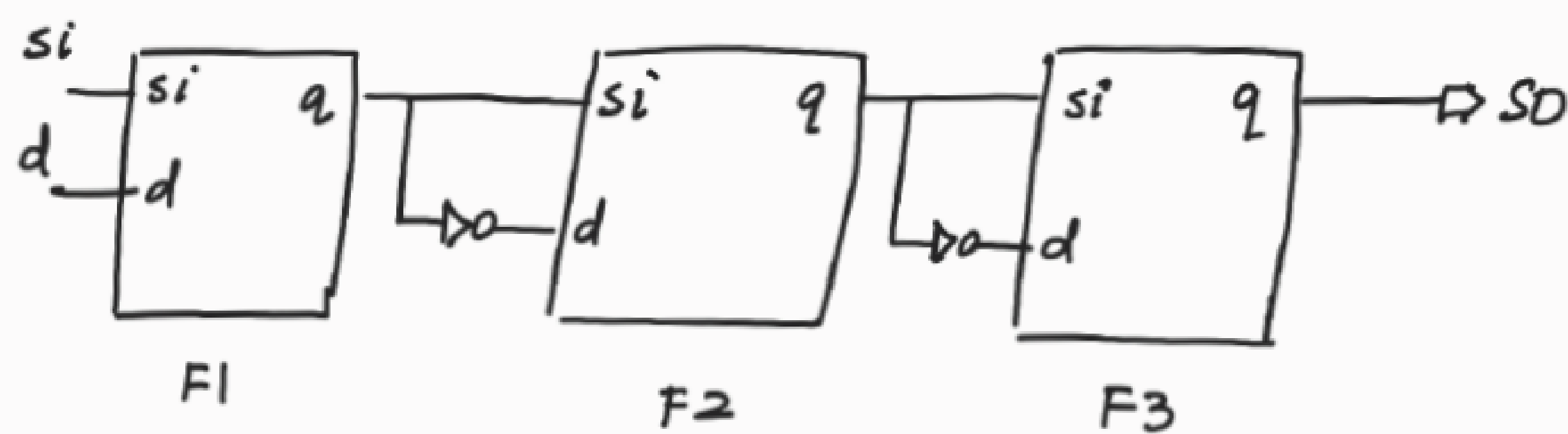


DIFFICULT INTERVIEW QUESTIONS IN SCAN INSERTION

Q1:

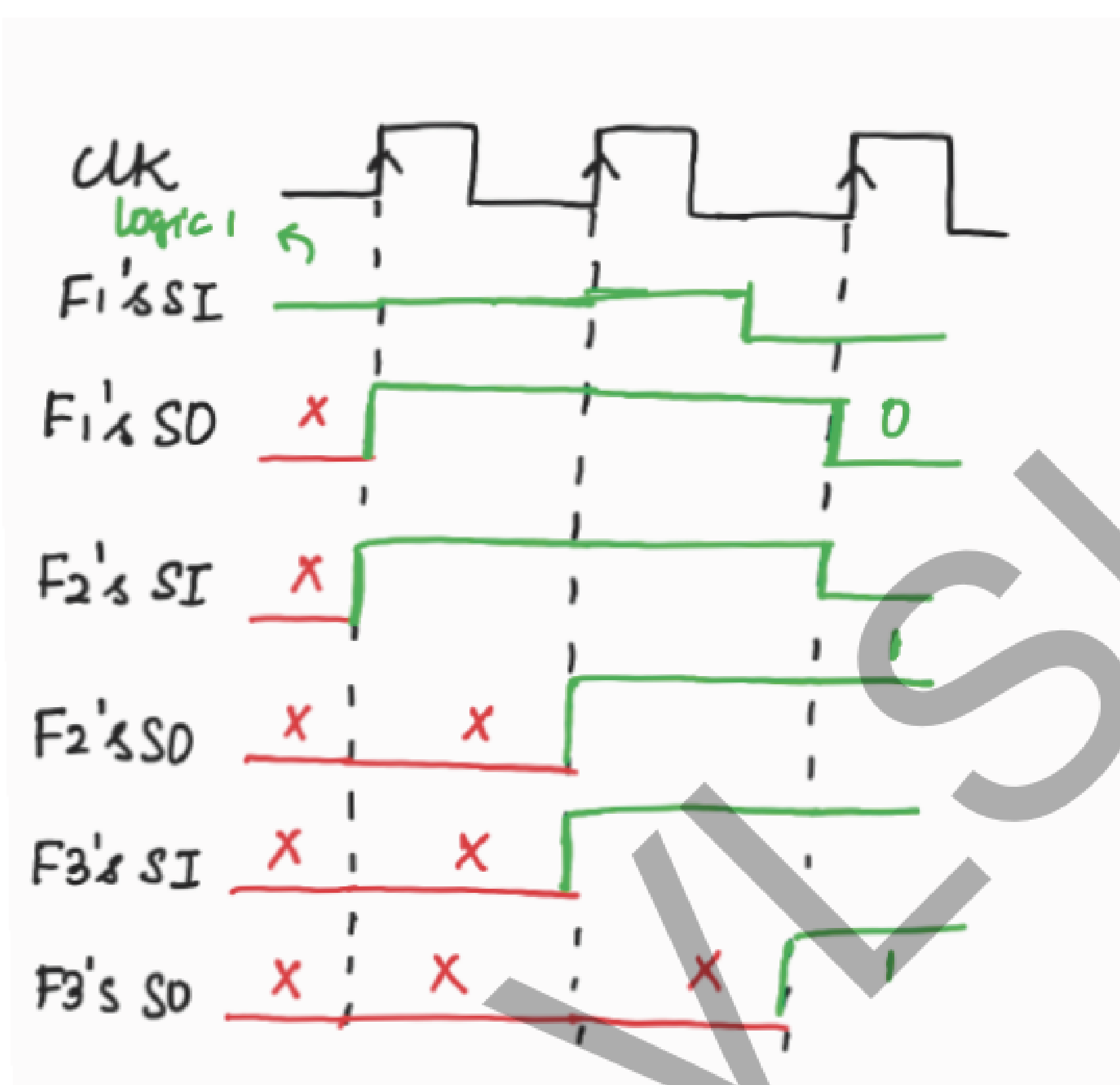
Q1: Explain scan chain shifting using waveform.



$F1's\ SD \Rightarrow F2's\ SI$

$F2's\ SD \Rightarrow F3's\ SI$

Pattern: 011



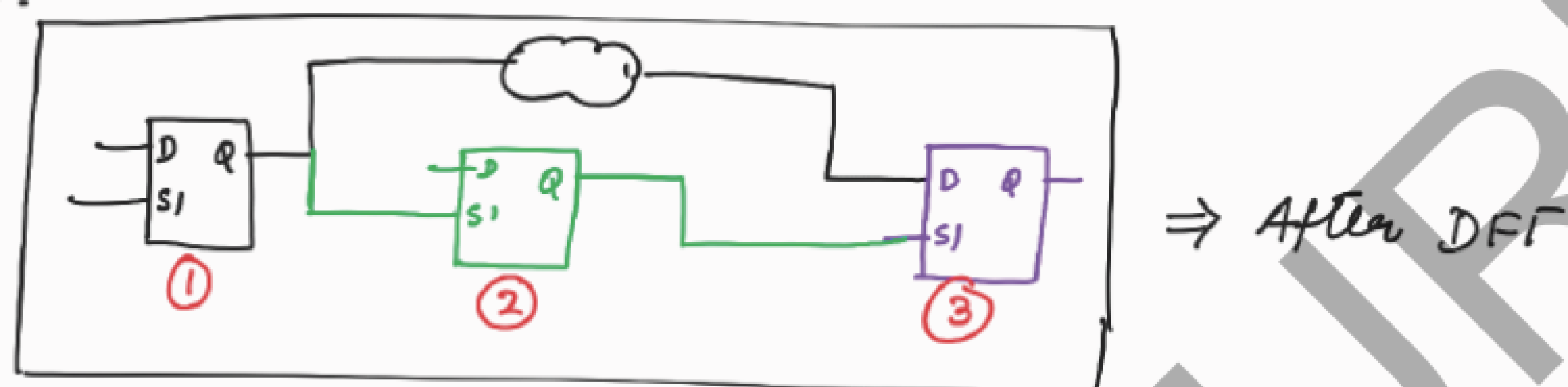
Q2:

Q5:- Why PD team require scanfile file? What will happen if scanfile is not provided?

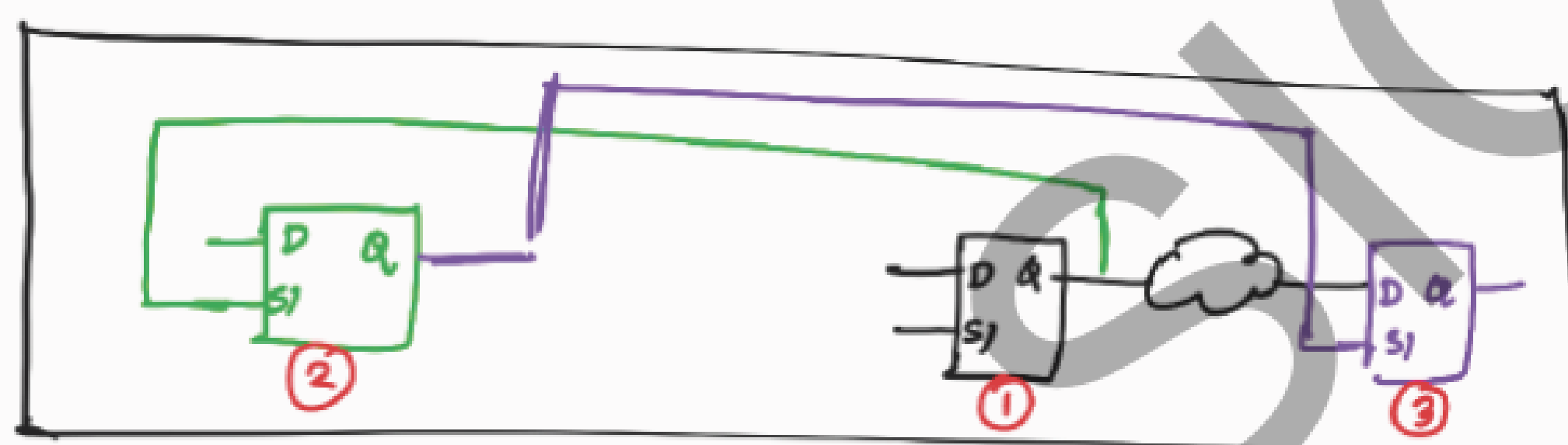
Ans:- For scan chain re-ordering, PD team need scanfile because that file only contain info about which cell can be re-ordered.

To avoid congestion, PD team has to do scan chain re-order.

NOTE:



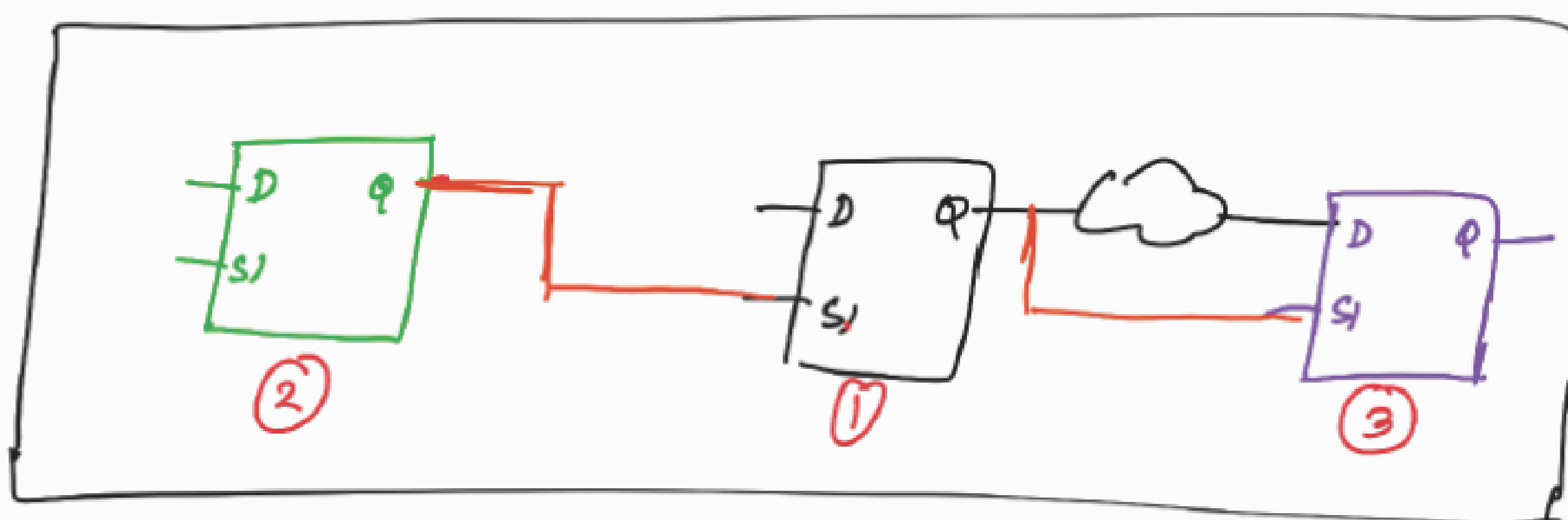
PD team
↓
(after placement)



If the original order of scan chain is followed:

- Zig-zag connectivity
- routing congestion
- wires will get shorted.

∴ DFT team can give flexibility to the PD team to re-order the scan chain.



After scan chain re-ordering.

If scanef file is not given to PD, then scan chain re-ordering can't be done and it will result in congestion.

Q3:

Q6: Why do we have non scan ffs in the design?

Ans:- Reason 1:-

If a flop is having clock controllability (or) set/reset controllability etc, and if it is not fixed, it will not be converted to scan ffs.

Reason 2 :-

In certain situations, the designer would have specified that these ffs should not be scanned:

Situation (i) :- The flop which hold some encryption data.

The designer asks us not to scan those ffs because if suppose it is scanned, the

encryption value will be shifted out. But it is a protected data. So those ffs will not be scanned.

situation(ii) :- The ffs that are used for security reasons.

Eg:- Some ffs will be used to lock JTAG.

Suppose company A is selling their chip to company B, company A will lock their JTAG and give.

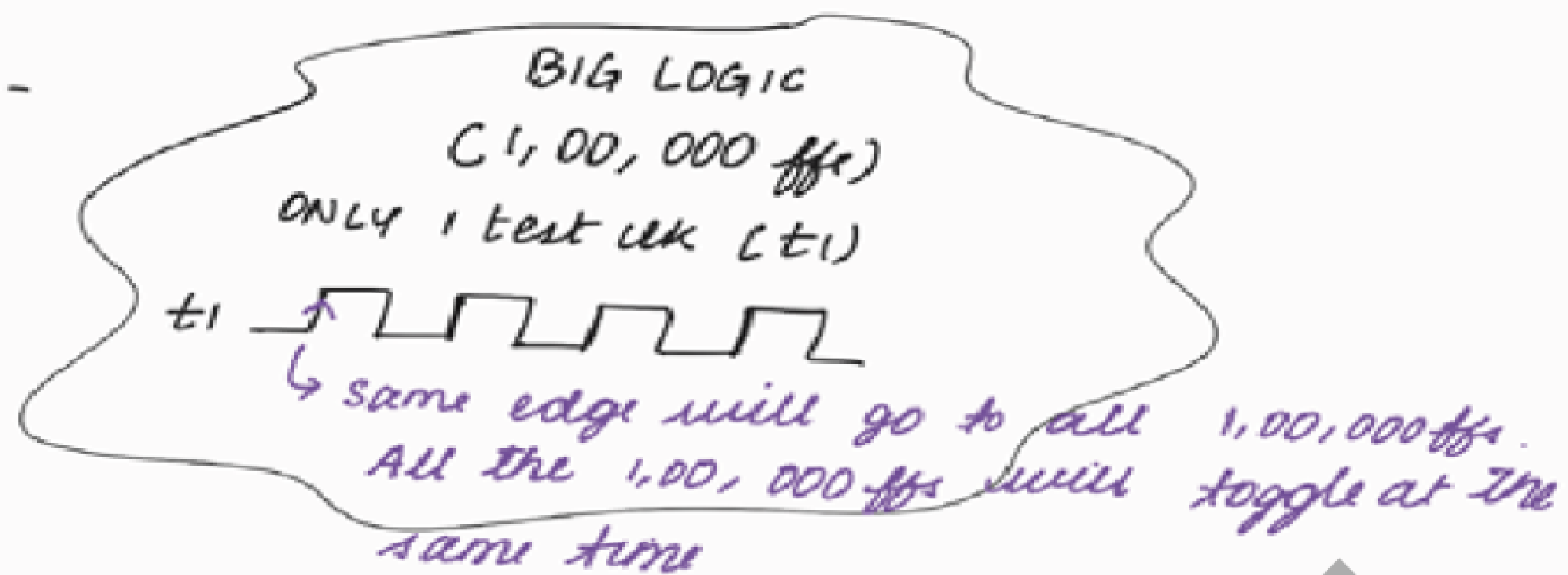
If suppose those security ffs are scanned, then there is a possibility that company B can shift some values into those ffs & remove the JTAG lock & company B will take the proprietary info of company A's chip.

So the security ffs will not be scanned & it will be designed in such a way that their clocks are also gated during scan operation so that their values can't be disturbed by anyone.

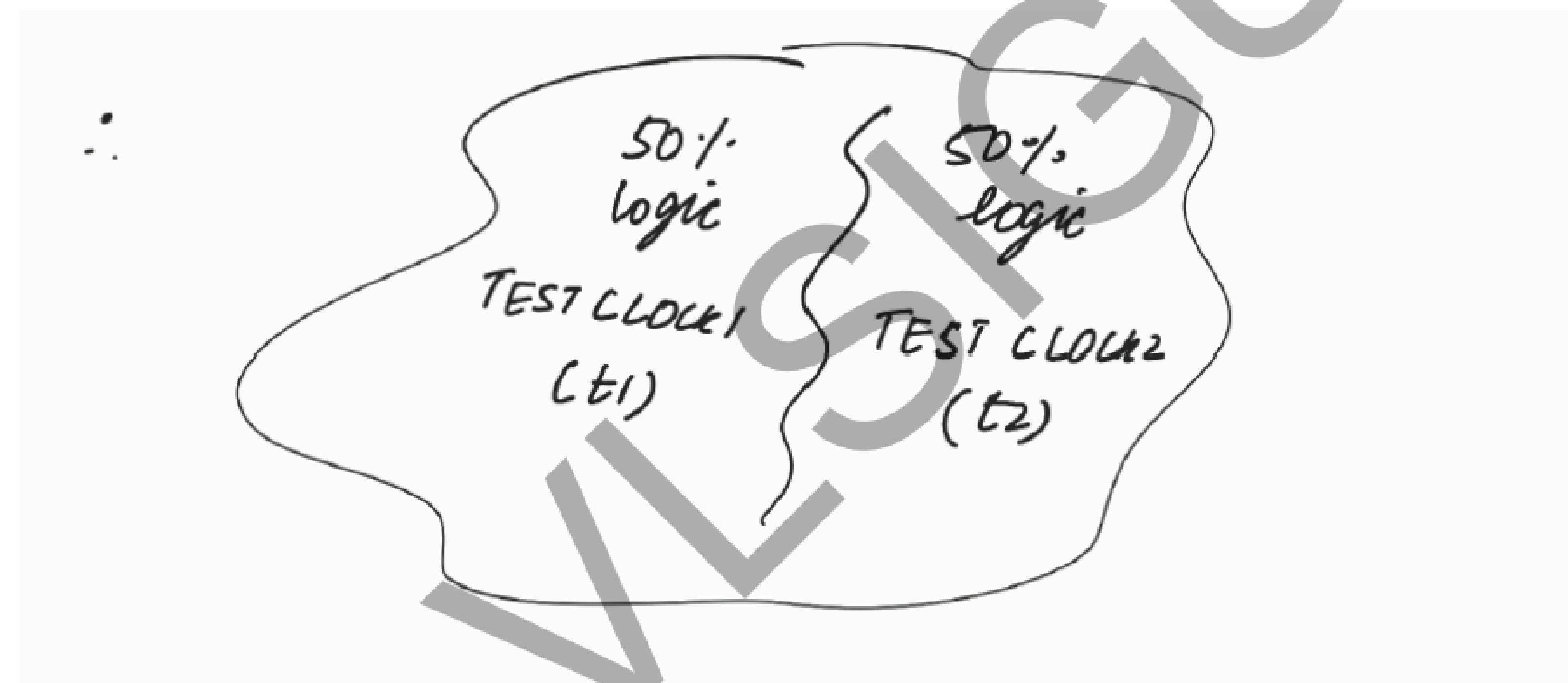
Q4:

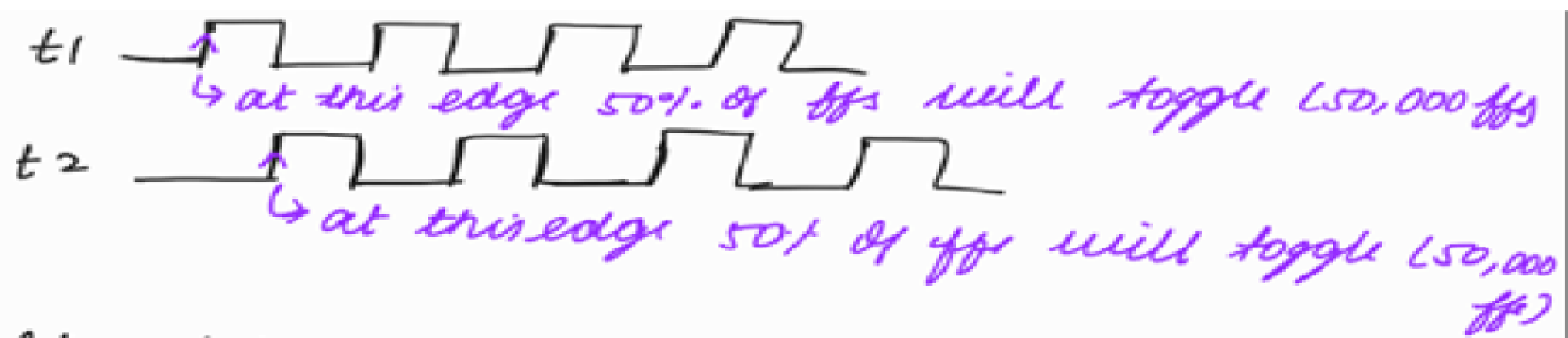
Q10:- Why do we need more than 1 scanclk (test clk) in the design?

Ans :-



↓
This might cause power issues in silicon





At a time, all 1,00,000 ffs are not toggling.
Only 50,000 ffs will toggle at a time.

$\therefore$ Power issues can be avoided.

Q5:

Q11:- Why scan insertion is performed on gate level netlist?

Ans:- In RTL, we don't have a physical impression of all flops.

RTL can also be written with parameters stating multiple ffs with single always command.

It is difficult to modify such code lines and mux to make it scan ff.

Q6:

Q: (DFT ques)

What steps we take while doing scan planning?

Ans:- We have to decide the following:

- (i) To check for which blocks comp is required
(if a block is very small, we don't require compression)
- (ii) no. of EDT channels + scan chains.
- (iii) shift frequency.